

# NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

**Subject**

Cybersecurity Law &  
Policy

**Project Type**

Group / Individual Project

**Total Marks** 100

Marks

**Deliverables**

Report + Presentation + Flowcharts +  
Infographics

**Important Notice to Students**

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

|                |                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
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## PART 1: Introduction

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Prepare an overview section that sets the foundation for your cybersecurity law. This section should clearly articulate why cybersecurity legislation is necessary for VIT's.

### What to Include

- Purpose of the law — What is this law designed to achieve?
- Importance of cybersecurity — Why does VIT's need this law now?
- Challenges faced by VIT's — Describe specific threats from the scenario
- Goals and expected outcomes — What will change after this law is enacted?

#### Scenario Context

VIT's is a rapidly growing digital nation with 50 million internet users, online banking, e-commerce platforms, government digital services, smart city infrastructure, AI-based public services, and cloud-based businesses. Recently, VIT's experienced large-scale data breaches, ransomware attacks on hospitals, banking fraud, social media misinformation campaigns, identity theft cases, and cyber bullying incidents.

### Overview

As VIT continues its transformation into a highly connected digital nation, cybersecurity has become a matter of nation security, economic stability, and public trust. With over 50 million internet users, widespread online banking, e-commerce services, smart city infrastructure, AI-driven public services, cloud computing, and digital government platforms, the nation increasingly relies on secure and resilient digital systems. However, the rapid expansion of technology has also exposed VIT to significant cyber threats that can disrupt critical services, compromise personal data and undermine confidence in the digital economy.

To address these growing risks, the Cybersecurity Law of VIT establishes a comprehensive legal framework for protecting digital infrastructure, securing personal and organization's data, preventing cybercrime, and promoting responsible use of technology across the country.

### **Purpose of the Law:**

The primary purpose of this law is to:

- Protect individuals, businesses and government institutions from cyber threats.
- Safeguard sensitive personal, financial and organizational data.
- Establish legal obligations for cybersecurity practices and incident reporting.
- Prevent, investigate and detect cybercrimes effectively.

- Strengthen the resilience of critical national infrastructure and digital services.
- Promote public confidence in the digital economy and online services.

## Importance of Cybersecurity:

VIT requires this law now because digital technologies have become essential to everyday life and national development. Citizens depends on online banking, healthcare systems, e-commerce platforms, social media networks, and government services. Any disruption to these systems can result in financial losses, privacy violations, service outages and threats to public safety.

Without a strong cybersecurity framework, cybercriminals can exploit vulnerabilities in digital systems, causing widespread harm to individuals, organizations and the nation as a whole.

## Challenges Faces by VIT :

Recent incidents have demonstrated the urgent need for cybersecurity legislation. Key challenges include:

1. Large-Scale Data Breaches
  - Unauthorized access to databases has exposed sensitive personal and financial information of citizens.
  - Organizations have suffered reputational damage and financial losses due to inadequate security controls.
2. Ransomware Attacks on Hospitals
  - Healthcare systems have been targeted by ransomware, disrupted medical services and putting patient safety at risk.
  - Critical health records have become inaccessible during emergencies.
3. Banking Fraud
  - Cybercriminals have exploited online banking systems through phishing, malware and fraudulent transactions.
  - Citizens and financial institutions have experienced significant financial losses.
4. Social Media Misinformation Campaigns
  - False and misleading information has spread rapidly across digital platforms.
  - Such campaigns threaten public trust, social stability and informed decision making.
5. Identity Theft
  - Criminals have stolen personal information to impersonate individuals and commit financial fraud.
  - Victims face economic harm and long term privacy concerns.
6. Cyberbullying and Online Harassment
  - Increasing cases of cyberbullying have negatively affected the mental well being and safety of users, particularly young people.
  - Existing legal mechanisms are insufficient to address harmful online behavior effectively.

## Goals and Expected Outcomes:

The Cybersecurity Law of VIT aims to achieve the following outcomes:

- Stronger protection of personal and organizational data.
- Reduced frequency and impact of cyberattacks through mandatory security standards.
- Faster detection, reporting and response to cybersecurity incidents.
- Enhanced protection of critical infrastructure, including healthcare, banking and government systems.
- Improved public awareness of cybersecurity risks and safe online practices.
- Greater accountability for organizations that handle sensitive information.
- More effective investigation and prosecution of cybercriminals.
- Increases trust in digital services, supporting economic growth and technological innovation.

## PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

### Required Terms to Define

| # | Term                  | Definition (To Be Completed by Student)                                                                                                             |
|---|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Cybercrime</b>     | Any unlawful act committed using computers, digital devices, networks or the internet that causes harm to individuals, organizations or the nation. |
| 2 | <b>Hacker</b>         | A person who gains access to computer systems or networks by exploiting vulnerabilities or security weaknesses.                                     |
| 3 | <b>Ethical Hacker</b> | A cybersecurity professional authorized to test and identify security weaknesses in systems to improve protection                                   |
| 4 | <b>Malware</b>        | Malicious software designed to damage, disrupt, spy on or gain unauthorized access to computer systems.                                             |
| 5 | <b>Ransomware</b>     | A type of malware that encrypts data or locks systems and demands payment for restoration of access.                                                |
| 6 | <b>Phishing</b>       | A fraudulent attempt to obtain sensitive information by impersonating a trusted entity through emails, messages, websites or other digital means.   |

|    |                                |                                                                                                                                                                                   |
|----|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | <b>Data Breach</b>             | An incident involving unauthorized access, disclosure, theft, alteration or destruction of data.                                                                                  |
| 8  | <b>Personal Data</b>           | Any information that can directly or indirectly identify an individual, including name, address, phone number, email or identification number.                                    |
| 9  | <b>Sensitive Data</b>          | Personal information required enhanced protection, including biometric data, health records, financial information and government issued identifiers                              |
| 10 | <b>Digital Signature</b>       | An electronic method of verifying the authenticity, integrity and origin of a digital document or transaction.                                                                    |
| 11 | <b>Electronic Evidence</b>     | Information stored or transmitted in digital form that may be used as evidence in legal or investigative proceedings.                                                             |
| 12 | <b>Identity Theft</b>          | The unauthorized acquisition, use or misuse of another person's personal information for fraudulent or criminal purposes.                                                         |
| 13 | <b>Artificial Intelligence</b> | Technology that enables computer systems to perform tasks that normally require human intelligence, such as learning, reasoning and decision making.                              |
| 14 | <b>Cloud Computing</b>         | The delivery of computer services, including storage, processing and software over the internet through remote servers.                                                           |
| 15 | <b>Critical Infrastructure</b> | Systems and assets essential to national security, public safety, economic stability and government operations, including banking, healthcare, energy and transportation systems. |
| 16 | <b>Cyberbullying</b>           | The use of digital platforms to harass, threaten, intimidate or humiliate individuals.                                                                                            |
| 17 | <b>Deepfake</b>                | Digitally generated or manipulated audio, video or images that falsely depict a person saying or doing something they did not actually say or do.                                 |
| 18 | <b>Encryption</b>              | The process of converting information into a coded format to prevent unauthorized access and ensure confidentiality.                                                              |
| 19 | <b>Firewall</b>                | A security system that monitors, filters and controls incoming and outgoing network traffic based on predefined security rules.                                                   |
| 20 | <b>Zero-Day Exploit</b>        | An attack that takes advantage of a software vulnerability before the developer has released a security patch or fix.                                                             |

## PART 3: Citizens' Digital Rights

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Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

### Required Rights

#### Right to Privacy

Citizens must be protected from unauthorized collection, use, or sharing of their personal data.

*[Student: Expand this right with specific provisions, who is responsible, and how it will be enforced.]*

Every citizen of VIT has the right to privacy in the digital environment. Personal information may only be collected, processed, stored or shared for lawful and clearly defined purposes. No organization, government agency or individual may access personal data without proper authorization or legal justification.

Organizations must obtain informed consent before collecting personal information. Privacy notices must clearly explain what data is being collected, why it is being collected, how long it will be retained and whether it will be shared with third parties. Special protection shall be provided for sensitive data such as health records, biometric information, financial details and government issued identification numbers.

This right shall be enforced by the Data Protection Authority (DPA) and the National Cyber Security Authority (NCSA). Organizations that unlawfully collect, misuse or disclose personal data may be subject to fines, suspension of operations and compensation claims for affected citizens.

#### Right to Secure Data

All organizations must store and protect citizens' data using industry-standard security measures.

*[Student: Expand this right with specific provisions, who is responsible, and how it will be enforced.]*

Every citizen has the right to expect that organizations handling their information will implement appropriate cybersecurity measures to protect it from unauthorized access, loss, theft, destruction or alteration. Data security is a fundamental responsibility of all entities operating within VIT.

Organizations must implement industry standard safeguards including encryption, multifactor authentication, access controls, network security systems, regular vulnerability assessments and secure backup mechanisms. Critical sectors such as healthcare, banking, cloud computing and government services shall be subject to enhanced security requirements.

The NCSA shall conduct cybersecurity audits and compliance inspections. Organizations found negligent in protecting personal data may face substantial penalties, mandatory remediation measures and public disclosure of violations.

## Right to Data Deletion

Citizens may request the complete deletion of their personal data from any organization's systems.

*[Student: Expand this right with specific provisions, who is responsible, and how it will be enforced.]*

Citizens have the right to request the deletion of personal data that is no longer necessary, has been unlawfully collected or is no longer required for the purpose for which it was obtained. This right helps citizens maintain control over their digital footprint.

Organizations must provide accessible mechanisms for submitting deletion requests through websites, mobile applications or customer service channels. Requests must be processed within thirty days unless a legal obligation requires continued retention of the data. When deletion is approved, information must be removed from active systems and databases.

The Data Protection Authority shall oversee compliance with deletion requests. Citizens may file complaints if organizations refuse legitimate requests. Failure to comply may result in administrative fines and corrective orders.

## Right to Information

Citizens have the right to know what data is collected about them and how it is used.

*[Student: Expand this right with specific provisions, who is responsible, and how it will be enforced.]*

Citizens have the right to know what information is collected about them and how it is used. Transparency is essential to maintaining trust between citizens and organizations operating in the digital economy.

Organizations must provide clear privacy policies and allow individuals to access copies of their personal data upon request. Citizens must be informed about data sharing practices, automated decision making systems, profiling activities and any third parties receiving their information.

The DPA may require organizations to disclose data processing activities and investigate complaints related to transparency violations. Organizations that fail to provide accurate information may be subject to regulatory sanctions and financial penalties.

## Right to Report Cybercrime

Citizens can report cybercrimes without fear of retaliation, and authorities must act within defined timeframes.

*[Student: Expand this right with specific provisions, who is responsible, and how it will be enforced.]*

Every citizen has the right to report cybercrime safely, confidentially and without fear of retaliation. Citizens should be able to seek assistance whenever they become victims of online fraud, cyberbullying, identity theft, hacking or other cyber offenses.

VIT shall establish a National Cybercrime Reporting Portal, a 24 hour cybercrime hotline, and regional cybercrime support centers. Citizens may submit reports anonymously when appropriate. Authorities must acknowledge reports within 24 hours and begin preliminary assessments within a legally defined timeframe.

Law enforcement agencies and the NCSA shall be responsible for investigating reports. Officials who fail to process complaints properly may face disciplinary measures. Whistleblowers reporting cybersecurity violations shall receive legal protection.

## Right to Fair Investigation

Any person accused of a cybercrime is entitled to a fair, transparent, and timely investigation.

*[Student: Expand this right with specific provisions, who is responsible, and how it will be enforced.]*

Any individual accused of a cybercrime has the right to a fair, impartial and transparent investigation. Cybersecurity laws must protect citizens while also ensuring that legal rights are respected during enforcement actions.

Investigators must collect digital evidence using lawful forensic procedures and maintain proper chains of custody. Individuals under investigation shall have the right to legal representation, access to relevant information and opportunities to challenge evidence presented against them.

Courts and independent oversight bodies shall monitor compliance with due process requirements. Evidence obtained through unlawful surveillance, unauthorized searches or improper investigative techniques may be excluded from legal proceedings.

## Instruction

You must include at least 6 rights. You may add more. Each right should be at least 2-3 paragraphs with specific implementation details.



## PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

| Crime                       | Description + Example                                                                                                                                     | Punishment                  | Fine                  |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------|
| <b>Unauthorized Access</b>  | Accessing a computer system, network, or account without permission.<br>Example: A hacker gains access to VIT's government citizen database.              | Up to 3 years imprisonment  | Up to 50,000rs Fine   |
| <b>Data Theft</b>           | Stealing confidential, personal or business data without authorization.<br>Example: A customer data stolen from an e-commerce platform.                   | Up to 5 years imprisonment  | Up to 1,00,000rs Fine |
| <b>Identity Theft</b>       | Using another person's personal information for fraud or impersonation.<br>Example: Criminal uses stolen national ID to open bank accounts.               | Up to 5 years imprisonment  | Up to 75,000rs Fine   |
| <b>Phishing</b>             | Deceiving users into revealing passwords or banking information through fake messages or websites.<br>Example: Fake bank emails targeting VIT's citizens. | Up to 4 years imprisonment  | Up to 60,000rs Fine   |
| <b>Ransomware</b>           | Encrypting systems or data and demanding payment for restoration.<br>Example: Hospital records locked by attackers.                                       | Up to 10 years imprisonment | Up to 5,00,000rs Fine |
| <b>Malware Distribution</b> | Creating or spreading malicious software that damages systems.<br>Example: Virus distributed through infected email attachments.                          | Up to 5 years imprisonment  | Up to 1,00,000rs Fine |

|                           |                                                                                                                                            |                            |                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------|
| <b>Cyberbullying</b>      | Using online platforms to harass, threaten, or intimidate individuals.<br>Example: Repeated online abuse of students on social media.      | Up to 2 years imprisonment | Up to 25,000rs Fine    |
| <b>Online Harassment</b>  | Sending threatening or abusive electronic communications.<br>Example: Persistent threats sent through messaging platforms.                 | Up to 3 years imprisonment | Up to 40,000rs Fine    |
| <b>Financial Fraud</b>    | Illegal online transactions or theft of money through digital means.<br>Example: Unauthorized transfer of funds from bank accounts.        | Up to 7 years imprisonment | Up to 2,00,000rs Fine  |
| <b>Misinformation</b>     | Deliberately spreading false information that causes public harm or panic.<br>Example: Fake emergency alerts circulated online.            | Up to 3 years imprisonment | Up to 50,000rs Fine    |
| <b>Website Defacement</b> | Unauthorized modification of website content.<br>Example: Government website homepage altered by attackers                                 | Up to 4 years imprisonment | Up to 75,000rs Fine    |
| <b>Cyber Terrorism</b>    | Cyberattacks targeting critical infrastructure or national security systems.<br>Example: Attack on VIT's power grid or smart city network. | Life Imprisonment          | Up to 10,00,000rs Fine |

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# VISUAL TAXONOMY OF CYBERCRIMES AND PENALTIES

Common Cyber Offenses, Real-World Examples in VIT's Context, and Applicable Punishments &amp; Fines

| #  | CYBERCRIME                  | DESCRIPTION + EXAMPLE<br>(within VIT's context)                                                                                                           | PUNISHMENT                  | FINANCIAL FINE        |
|----|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------|
| 1  | <b>UNAUTHORIZED ACCESS</b>  | Accessing a computer system, network, or account without permission.<br>Example: A hacker gains access to VIT's government citizen database.              | Up to 3 years imprisonment  | Up to ₹50,000 Fine    |
| 2  | <b>DATA THEFT</b>           | Stealing confidential, personal or business data without authorization.<br>Example: Customer data stolen from an e-commerce platform.                     | Up to 5 years imprisonment  | Up to ₹1,00,000 Fine  |
| 3  | <b>IDENTITY THEFT</b>       | Using another person's personal information for fraud or impersonation.<br>Example: Criminal uses stolen national ID to open bank accounts.               | Up to 5 years imprisonment  | Up to ₹75,000 Fine    |
| 4  | <b>PHISHING</b>             | Deceiving users into revealing passwords or banking information through fake messages or websites.<br>Example: Fake bank emails targeting VIT's citizens. | Up to 4 years imprisonment  | Up to ₹60,000 Fine    |
| 5  | <b>RANSOMWARE</b>           | Encrypting systems or data and demanding payment for restoration.<br>Example: Hospital records locked by attackers.                                       | Up to 10 years imprisonment | Up to ₹5,00,000 Fine  |
| 6  | <b>MALWARE DISTRIBUTION</b> | Creating or spreading malicious software that damages systems.<br>Example: Virus distributed through infected email attachments.                          | Up to 5 years imprisonment  | Up to ₹1,00,000 Fine  |
| 7  | <b>CYBERBULLYING</b>        | Using online platforms to harass, threaten, or intimidate individuals.<br>Example: Repeated online abuse of students on social media.                     | Up to 2 years imprisonment  | Up to ₹25,000 Fine    |
| 8  | <b>ONLINE HARASSMENT</b>    | Sending threatening or abusive electronic communications.<br>Example: Persistent threats sent through messaging platforms.                                | Up to 3 years imprisonment  | Up to ₹40,000 Fine    |
| 9  | <b>FINANCIAL FRAUD</b>      | Illegal online transactions or theft of money through digital means.<br>Example: Unauthorized transfer of funds from bank accounts.                       | Up to 7 years imprisonment  | Up to ₹2,00,000 Fine  |
| 10 | <b>MISINFORMATION</b>       | Deliberately spreading false information that causes public harm or panic.<br>Example: Fake emergency alerts circulated online.                           | Up to 3 years imprisonment  | Up to ₹50,000 Fine    |
| 11 | <b>WEBSITE DEFACEMENT</b>   | Unauthorized modification of website content.<br>Example: Government website homepage altered by attackers.                                               | Up to 4 years imprisonment  | Up to ₹75,000 Fine    |
| 12 | <b>CYBER TERRORISM</b>      | Cyberattacks targeting critical infrastructure or national security systems.<br>Example: Attack on VIT's power grid or smart city network.                | Life Imprisonment           | Up to ₹10,00,000 Fine |

## PART 5: Data Protection Framework

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Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

### Data Collection Rules

*[Student: Write detailed rules for this section. Include specific technical and legal requirements.]*

- Organizations may collect only data necessary for a legitimate purpose.
- Clear notice must be provided before data collection.
- Personal data cannot be collected through deceptive means.
- Consent must be obtained before collecting sensitive personal information.

### Data Storage Rules

*[Student: Write detailed rules for this section. Include specific technical and legal requirements.]*

- Personal Data must be stored using encryption.
- Organizations must implement access controls and authentication mechanisms.
- Regular security audit must be conducted.
- Backup systems must be maintained securely.

### Data Sharing Rules

*[Student: Write detailed rules for this section. Include specific technical and legal requirements.]*

- Data may only be shared for authorized purposes.
- Third parties must provide equivalent data protection standards.
- Citizens must be informed when their data is shared.
- Unauthorized sale of personal data is prohibited.

### User Consent Requirements

*[Student: Write detailed rules for this section. Include specific technical and legal requirements.]*

- Consent must be Freely given, informed and specific.
- Users may withdraw consent at any time.
- Separate consent is required for marketing activities.
- Consent records must be maintained by organizations.

## Data Retention Period

*[Student: Write detailed rules for this section. Include specific technical and legal requirements.]*

- Data shall be retained only as long as necessary.
- Financial records may be securely deleted.
- Unused personal data must be securely deleted.
- Retention schedules must be publicly available.

## Data Deletion Process

*[Student: Write detailed rules for this section. Include specific technical and legal requirements.]*

- Citizens may request deletion of unnecessary personal data.
- Organizations must respond within 30 days.
- Deleted data must be permanently removed from active systems.
- Exceptions apply where retention is required by law.

## Breach Notification Requirements

*[Student: Write detailed rules for this section. Include specific technical and legal requirements.]*

- Data breaches must be reported within 72 hours.
- Affected individuals must be informed without undue delay.
- Organizations must describe the nature and impact of the breach.
- Corrective actions must be reported to the Cybersecurity Authority.

# DATA PROTECTION & CORPORATE SECURITY FRAMEWORK

A Comprehensive Framework for All Organizations Operating in VIT's



## OUR COMMITMENT

Protect personal data, ensure privacy, maintain trust, and comply with the laws of VIT's.



Lawful & Transparent



Secure by Design



User Empowerment



Accountability & Compliance



### 1. DATA COLLECTION RULES

- Collect only data necessary for a legitimate purpose.
- Clear notice must be provided before data collection.
- Personal data cannot be collected through deceptive means.
- Consent must be obtained before collecting sensitive personal information.



### 2. DATA STORAGE RULES

- Personal data must be stored using encryption.
- Implement access controls and authentication mechanisms.
- Regular security audits must be conducted.
- Backup systems must be maintained securely.



### 3. DATA SHARING RULES

- Data may only be shared for authorized purposes.
- Third parties must provide equivalent data protection standards.
- Citizens must be informed when their data is shared.
- Unauthorized sale of personal data is prohibited.



### 7. BREACH NOTIFICATION REQUIREMENTS

- Data breaches must be reported within 72 hours.



- Affected individuals must be informed without undue delay.



- Organizations must describe the nature and impact of the breach.



- Corrective actions must be reported to the Cybersecurity Authority.



### COMPLIANCE & ENFORCEMENT

Organizations must comply with this framework and all applicable data protection laws in VIT's. Non-compliance may result in penalties, sanctions, and legal action.



### OVERSIGHT AUTHORITIES

- Data Protection Authority (DPA)
- National Cybersecurity Authority (NCSA)

## THE DATA LIFECYCLE

Responsible Handling at Every Stage



### 4. USER CONSENT REQUIREMENTS

- Consent must be freely given, informed and specific.
- Users may withdraw consent at any time.
- Separate consent is required for marketing activities.
- Consent records must be maintained by organizations.



### 5. DATA RETENTION PERIOD

- Data shall be retained only as long as necessary.
- Financial records may be securely deleted.
- Unused personal data must be securely deleted.
- Retention schedules must be publicly available.



### 6. DATA DELETION PROCESS

- Citizens may request deletion of unnecessary personal data.
- Organizations must respond within 30 days.
- Deleted data must be permanently removed from active systems.
- Exceptions apply where retention is required by law.

## PART 6: Corporate Responsibilities

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All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

### Required Responsibilities (Minimum 10)

#### Responsibility 1: Cybersecurity Policy

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Every organization must maintain a documented cybersecurity policy.
- Non-compliance may result in regulatory penalties.

#### Responsibility 2: Employee Security Training

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Annual cybersecurity awareness training is mandatory for all employees.
- Failure to conduct training may result in fines.

#### Responsibility 3: Risk Assessment

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Organizations must perform cybersecurity risk assessments annually.
- Critical vulnerabilities must be addressed promptly.

#### Responsibility 4: Security Audits

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Independent security audits must be conducted at least once every year.
- Audit reports must be submitted to regulators upon request.

## Responsibility 5: Multi-Factor Authentication (MFA)

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- MFA must be implemented for critical systems and administrator accounts.
- Failure to implement MFA must attract penalties.

## Responsibility 6: Data Encryption

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Sensitive personal and financial data must be encrypted both in storage and transmission.
- Unencrypted storage of critical data is prohibited.

## Responsibility 7: Incident Response Plan

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Organizations must maintain a documented incident response plan.
- Cyber incidents must be reported immediately.

## Responsibility 8: Data Protection Officer (DPO)

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Large Organization must appoint a qualified DPO.
- The DPO is responsible for privacy compliance and breach management.

## Responsibility 9: Vulnerability Management

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Security patches and updates must be applied regularly.
- Known vulnerabilities must not remain unaddressed.

## Responsibility 10: Third-Party Vendor Security

*[Student: Describe the specific obligation, who it applies to, and the consequence for noncompliance.]*

- Organizations are responsible for ensuring vendors meet cybersecurity requirements.
- Contracts must include cybersecurity obligations.

### Examples to Consider

Maintain security controls | Conduct regular security audits | Train employees on cybersecurity |

Report incidents within 72 hours | Protect customer information | Appoint a Data Protection Officer  
| Conduct annual penetration testing | Implement access control policies | Encrypt sensitive data |  
Maintain incident response plans

## PART 7: Government Responsibilities

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Specify how government agencies of VIT's must protect national interests in cyberspace. Include concrete mechanisms for each area.

### Protect National Infrastructure

*[Hint: Define which infrastructure is critical and how it must be protected.]*

Responsibility: Government shall identify and protect Critical Information Infrastructure (CII) which include Banking, Hospital, Power, Transportation, Government Database, Smart city, Telecommunication networks etc.

Applies to: Government agencies and operators of critical infrastructure.

Consequences of non-compliance: fines, enforcement measures, and license revocation for violations.

### Monitor Cyber Threats

*[Hint: Describe surveillance systems, threat intelligence gathering, and monitoring protocols.]*

- Task: Create a National Cyber Threat Monitoring Center to provide continuous monitoring of cyber threats, gather cyber threat intelligence, identify cyber attacks and provide cyber security alerts.
- Applies to: National Cyber Security Authority (NCSA) and relevant government agencies.
- Consequences for noncompliance: Internal audit and accountability process of responsible officials.

### Investigate Cybercrimes

*[Hint: Outline the legal processes for cyber investigations including warrants and digital forensics.]*

- Responsibility: Providing special units for cybercrime investigations, with the necessary digital forensic laboratories, to investigate cybercriminal offences and to trace digital evidence.
- For whom it is used: Law enforcement agencies, cybercrime investigators.
- Evidence handling procedures: If evidence handling procedures are not adhered to, investigations may be declared invalid.

## International Coordination

*[Hint: Explain how VIT's will work with other countries' cyber agencies and INTERPOL.]*

- Responsibility: Share information and intelligence with foreign governments, INTERPOL, CERTs, and international cybersecurity entities to help in the fight against cross-border cybercrime.
- It is applicable to the following: Ministry of Foreign Affairs, NCSA, and police and security forces.
- Consequences for failure to comply: Less capacity to investigate foreign cyber threats and recovery of stolen assets.

## Awareness Programs

*[Hint: Describe government-led public education campaigns about cybersecurity.]*

- Accountability: Promote digital literacy and cybersecurity education in schools and universities, and hold cybersecurity awareness campaigns throughout the country.
- Applies to: Ministry of Education, NCSA, public agencies.
- Consequences of non-compliance: Annual review/corrective action plans.

## PART 8: Cyber Incident Reporting Framework

Design a step-by-step process for how cyber incidents are reported, investigated, and resolved in VIT's. You must also create a flowchart (as a separate deliverable).

### Reporting Process Steps

| Step | Stage                  | Description (Student to Complete)                                                                                                                                              |
|------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Incident Detected      | Suspects activity or data breach, malware or system compromise is identified by the organization. Initial evidence is kept.                                                    |
| 2    | Internal Investigation | Security team determines the scope, severity, systems impacted, and potential impact of the incident.                                                                          |
| 3    | Report to Authority    | Any critical incident must be reported to the National Cyber Security Authority within 72 hours.                                                                               |
| 4    | Risk Assessment        | The authorities conduct a risk assessment, consisting of the determination of the threat level, potential damages, impact on citizens, businesses, or critical infrastructure. |
| 5    | Mitigation             | Immediate containment actions are undertaken, such as isolating systems, patching and removal of threats.                                                                      |
| 6    | Recovery               | Systems are restored from backups, services are up and running and security is enhanced.                                                                                       |
| 7    | Final Report           | Organisation reports back the final incident report, causes, impact, response actions and lessons learned.                                                                     |

#### Deliverable Reminder

You must submit a separate FLOWCHART showing this complete process. Use tools like Canva, draw.io, Lucidchart, or PowerPoint SmartArt to create a professional diagram.

## PART 9: National Cyber Security Authority (NCSA)

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Design the organizational structure, responsibilities, and powers of the NCSA — the primary government agency responsible for cybersecurity in VIT's.

### Organizational Structure

Incidents must be reported within 72 hours of the incident. Any incidents involving the hospital, bank and government systems need to be reported straight away. Penalties can be up to 200,000 VIT Dollars for non-reporting. It's illegal to lie.

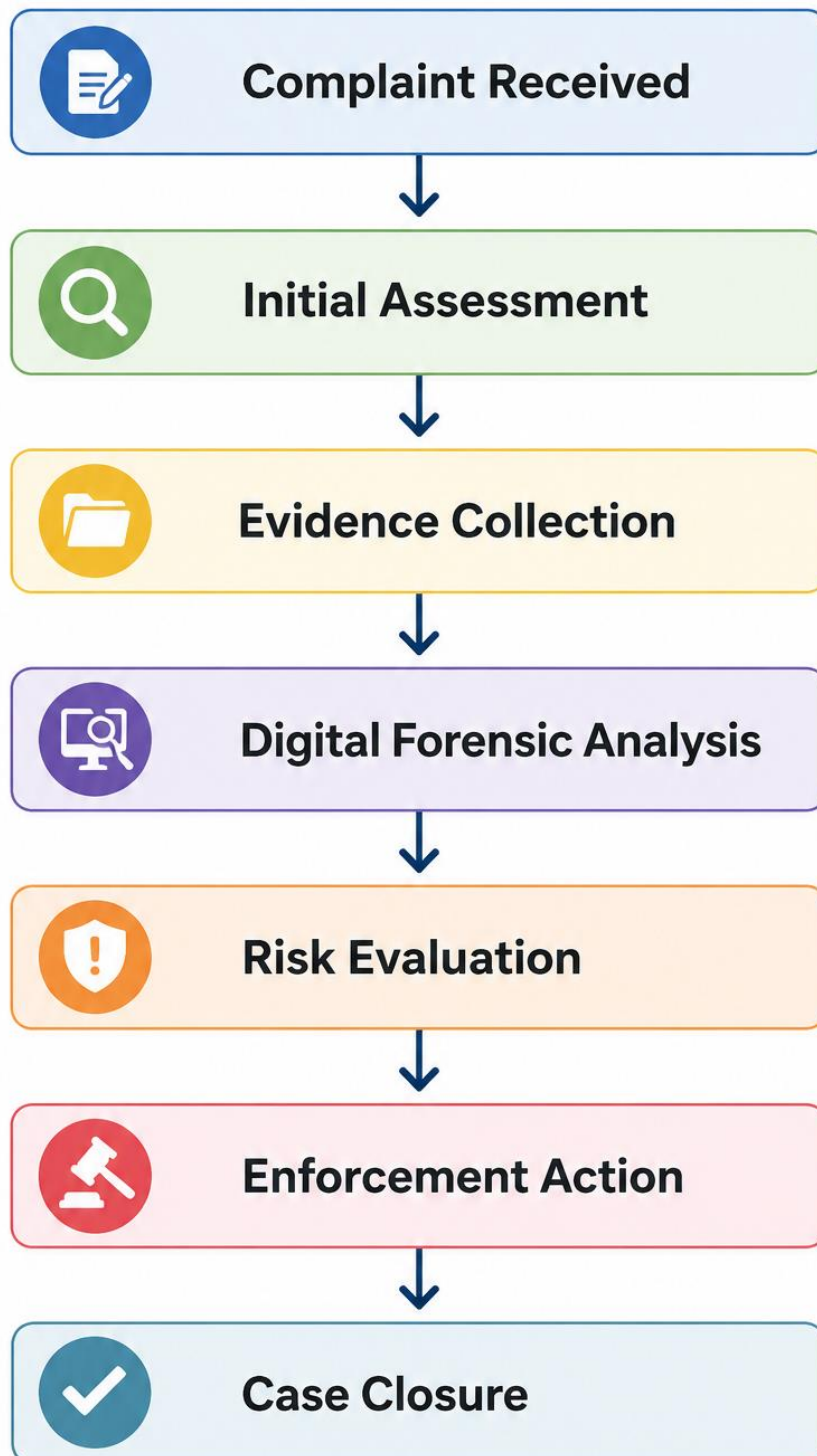
### Core Responsibilities

- Formulate National Cyber security policies.
- Track cyber issues and incidents.
- Protect critical infrastructure.
- Organize the national defence activities in the cyber space.
- Run cyber security awareness campaigns.
- Provide cybersecurity advisories and alerts.
- Support cybercrime investigations.
- Regulate cybersecurity compliance.

### Legal Powers

- Carry out Cyber Security Inspections/Audits.
- Ensure organizations report cyber incidents.
- Issue compliance directives.
- Impose administrative penalties.
- Implement emergency cyber countermeasures in national cyber crisis situations.
- Seek data pertinent to cyber investigations.
- Communicate and liaise with law enforcement.

## Investigation Process



## Departments

1. Threat Intelligence Department Takes measures to identify new cyber risks. Shares threat information on a sector basis.
2. Incident Response Department Leads response to significant cyber incidents. Gives emergency technical assistance.
3. Digital Forensics Laboratory Gathers and investigates electronic evidence. Supports criminal investigations.
4. Compliance & Audit Department Performs audit and inspections in computer security. Ensures adherence to cybersecurity legislation.
5. The 5th Department is Public Awareness & Education. Public Awareness & Education is the 5th Department. Conducts cybersecurity awareness programs. Creates training materials/educational programs.

## PART 10: AI and Emerging Technology Regulations

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Create specific cybersecurity regulations for emerging technologies used in VIT's. For each technology, identify key risks and write enforceable rules.

### Artificial Intelligence

#### Key Risks

- AI-generated misinformation and fake news.
- Decision making that is biased or discriminatory.
- Harvesting of personal information without permission.
- AI-driven phishing and hacking. Phishing and hacking by AI. Incomplete understanding of automated decision-making process

#### Regulations

- Risky AI systems should be subjected to security and ethical evaluations.
- Organizations need to publicize when AI is making major decisions.
- Data protection laws must be adhered to by the AI systems that are collecting personal information.
- Any content created using AI should be properly marked. Businesses continue to be liable for damage from their AI systems.

### Deepfakes

#### Key Risks

- Identity fraud
- Election manipulation
- Reputation damage
- Fake videos/Voicemails (financial scams.)
- Public misinformation campaigns

#### Regulations

- The creation of fake deepfakes is not allowed.
- Digital Watermarking is required on AI-generated media.
- Victims are entitled to ask to be taken away right away.
- Platforms are expected to take down harmful deepfakes when they are notified of them.
- Deepfakes for fraud and extortion are subject to criminal penalties.

## IoT Devices

### Key Risks

- Weak passwords
- Device hijacking
- Botnet attacks
- Data leakage
- Smart home surveillance threats

### Regulations

- The default passwords should be changed when setting up.
- Security patches need to be offered by the manufacturers.
- It is important that IoT devices be able to support encryption.
- Report and fix security bugs.
- Certification needs to be done prior to widespread deployment.

## Cloud Computing

### Key Risks

- Data breaches
- Insider threats
- Misconfigured cloud storage
- Service disruptions
- Cross-border data exposure

### Regulations

- Cloud providers need to provide encryption.
- It is necessary to keep critical government data in a designated area.
- Security audits are required on a regular basis.
- Incident must be reported within 72 hours.
- Access and monitoring of the system must be tight.

## Digital Identity Systems

### Key Risks

- Identity theft
- Biometric data theft
- Unauthorized access
- Fraudulent registrations
- Database compromise

### Regulations

- Multi-factor authentication required.
- Biometric data should be encrypted.
- Citizens should have access to and the right to correct their identity records.
- Digital identities can be used for unauthorized purposes and will be punished.
- Regular assessments of security are required.

## Autonomous Systems

### Key Risks

- System malfunction
- Remote hijacking
- AI decision errors
- Safety risks
- Data manipulation

### Regulations

- Required testing prior to deployment.
- Ongoing AS monitoring.
- Emergency shutdown mechanisms required.
- Security patches need to be applied on a regular basis.
- Operators are still liable for the use of the system.

## PART 11: Critical Infrastructure Protection

Create sector-specific cybersecurity requirements for each of the following critical infrastructure types in VIT's. Requirements should be practical and technically grounded.

| Sector                            | Cybersecurity Requirements (Student to Complete)                                                                                                        |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Banks &amp; Financial</b>      | Multi Factor Authentication (MFA), encryption of financial information, 24/7 monitoring, annual penetration testing, incident reporting within 24 hours |
| <b>Hospitals &amp; Healthcare</b> | Security of patient records, ransomware protection strategies, secure medical devices, frequent backups, employee cybersecurity training                |
| <b>Airports &amp; Aviation</b>    | Network segmentation, access control systems, disaster recovery plans, security audits, threat monitoring                                               |
| <b>Railways &amp; Transport</b>   | Real-time monitoring and secure communication systems, incident response plans, vulnerability assessments, backup control systems                       |
| <b>Power Plants &amp; Energy</b>  | Industrial Control System (ICS) security, network isolation, threat intelligence sharing, continuous monitoring and emergency procedures for recovery.  |

**Government Data Centers**

Data encryption, access management, security audits, backup facilities, mandatory compliance reviews – all of these are provided by Government Data Centers.

## PART 12: Awareness and Education

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Design a National Cybersecurity Awareness Campaign for VIT's. Your campaign must reach all segments of society through age-appropriate and effective programs.

### School Programs (Age 6-18)

**Topic:**

Safe internet use, Password security, Cyberbullying awareness, Social media safety, Online privacy,

**Activities:**

Cybersecurity awareness week, Interactive workshops ,Online safety quizzes ,Poster competitions

**Learning Outcomes**

Recognize online threats ,Follow safety rules on the Internet.,Report cyberbullying incidents

### College & University Programs

**Topics:**

Cybersecurity fundamentals, Ethical hacking, Digital forensics, Secure software development  
Data privacy laws.

**Requirements:**

Cybersecurity laboratory sessions ,Industry certifications, Internship opportunities, Practical security projects

## Corporate Training Programs

**Training Requirements:**

Annual mandatory training, Phishing simulation exercises, Incident reporting procedures,  
Password management training ,Data protection compliance training.

**Assessment:**

Online tests, Security awareness evaluations ,Compliance certification

## Social Media Campaigns

**Platforms:**

Instagram ,X (Twitter), LinkedIn ,YouTube ,Facebook

**Campaign Messages:**

If you aren't sure, think before you click! Protecting your Digital Identity, Report Cybercrime,  
Stay Safe Online, Content Types, Short videos, Infographics, Awareness posters,Cybersecurity tips

## Public Awareness Workshops

**Locations:**

Community centers, Schools, Universities, Government offices

**Target Audience:**

Students, Senior citizens, Small business owners, Government employees

**Topics:**

Online fraud prevention, Digital payments security, Data privacy, Cybercrime reporting procedures

## PART 13: International Cooperation

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Define how VIT's will collaborate with other countries and international organizations to combat cybercrime that crosses national borders.

### Cybercrime Investigations

**Policy:**

VIT can collaborate with foreign law enforcement to investigate cybercrime transnational offenses. Digital evidence can be exchanged via the legal system and exchange of assistance. Investigations can be shared between agencies for cyber incidents involving a large number of people.

**Example:**

A ransomware group targets hospitals in VIT's but is based in another nation. VIT gets in touch with foreign cybercrime units for identification and apprehension of the attackers.

### Information Sharing Agreements

**Policy:**

VIT's can participate in information exchange agreements with trusted countries and international bodies in the area of cybersecurity. Threat intelligence, malware characteristics and attack patterns can be shared. Information that is shared must be kept secret and not disclosed to others.

**Example:**

Before the new banking malware attack can reach VIT's banking institutions, the foreign CERT alerts VIT's about the attack.

## Joint Operations

**Policy:**

VIT can be involved in global cybercrime activities. National agencies can associate with INTERPOL, CERTs and cyber defense groups. Joint exercises can be held to enhance preparedness.

**Example:**

VIT's joins global effort to break-up phishing network targeting online banking consumers.

## Extradition Requests

**Policy:**

Persons charged with serious criminal activities on the Internet can be extradited under international conventions. Extradition requests must be based on due process and human rights protections. Extradition may be sought in the case of cyber terrorism and the large scale financial cybercrime.

**Example:**

The cybercriminal, whose name has been kept secret, who stole millions from VIT's banks is arrested in a foreign country and extradited for prosecution.

## Threat Intelligence Exchange

**Policy:**

The National Cyber Security Authority (NCSA) shall provide Cyber threat Intelligence to other countries. IOCs, malware signatures and threat reports can be shared. Information sharing shall be done in a secure and legal manner.

**Example:**

VIT's gets notified of a new ransomware that it can then prepare its defenses against.

## PART 14: Case Studies

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Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

### Case Study 1: Government Citizen Database Breach

|                              |                                                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Crime Category</b>        | Unauthorized Access / Data Breach                                                                                               |
| <b>What Happened</b>         | Access to personal data of 500,000 citizens was gained by exploiting an unpatched vulnerability on a government database.       |
| <b>Law Violated</b>          | The law that was violated is Unauthorized Access, Data Theft, Privacy Violations.                                               |
| <b>Investigation Process</b> | NCSA was able to detect the breach, gather digital evidence, track back to the attack's origins, and work with law enforcement. |
| <b>Punishment Given</b>      | 6 years imprisonment and 250,000 VIT Dollars fine.                                                                              |

|                        |                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------|
| <b>Lessons Learned</b> | Introduction of mandatory patch management, increased access control, and regular security audit. |
|------------------------|---------------------------------------------------------------------------------------------------|

## Case Study 2: National Bank Phishing Scam

|                              |                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Crime Category</b>        | Financial Fraud / Phishing                                                                                                                  |
| <b>What Happened</b>         | The criminals also had a fake banking site and offered phishing emails to thousands of banking customers. Hundreds of accounts were hacked. |
| <b>Law Violated</b>          | Phishing, Identity Theft, Financial Fraud                                                                                                   |
| <b>Investigation Process</b> | The NCSA investigated phishing infrastructure, retrieved phishing domains, and assisted banks in determining victims.                       |
| <b>Punishment Given</b>      | 8 years imprisonment and 400,000 VIT Dollars fine.                                                                                          |
| <b>Lessons Learned</b>       | Mandatory MFA, Customer awareness campaigns and enhanced email security controls.                                                           |

## Case Study 3: Hospital Ransomware Attack

|                              |                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Crime Category</b>        | Ransomware / Cyber Terrorism                                                                                         |
| <b>What Happened</b>         | Attackers were able to encrypt patient data in several hospitals, and thus interrupt the emergency medical services. |
| <b>Law Violated</b>          | Ransomware Distribution, Critical Infrastructure Attack                                                              |
| <b>Investigation Process</b> | NCSA responded to incidents, conducted digital forensics and traced cryptocurrency payments.                         |
| <b>Punishment Given</b>      | 12 years imprisonment and 750,000 VIT Dollars fine.                                                                  |
| <b>Lessons Learned</b>       | Backup Offline, Segmentation, Incident Response Planning, and Employee Training were required.                       |

## PART 15: Future Recommendations

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Based on your research, suggest forward-looking improvements and additions to VIT's cybersecurity law for the next 10 years. Consider how technology will evolve and what new threats will emerge.

### AI Security (2025-2035)

#### Recommendations:

- AI risk assessments will be required.
- Control of high-risk AI systems.
- Requirements for AI-generated content—labelling.
- 3rd-party audits of government AI systems.
- A national board for AI Ethics and Security.

### Quantum Computing

#### Recommendations:

- Launch migration to quantum resistant cryptography.
- Found a national quantum security research institution.
- Press critical infrastructure operators to make a quantum risk assessment.
- Create a country-wide quantum preparedness roadmap.

### Privacy Protection

#### Recommendations:

- Improved safeguards for biometric information.
- Improved behavioral profiling controls.
- Requirements for privacy in digital services.
- Public awareness of automated decision-making.

### Digital Identity

#### Recommendations:

- National secure digital identity system.
- Decentralized identities for verification.
- Biometric encryption standards.
- Control of identity information by citizens.

## Cyber Warfare

### Recommendations:

- Describe cyber warfare and cyber aggression as per the national law.
- Prepare a countrywide cyber defense policy.
- Create a Cyber Defense Command.
- Develop strategies to defend against state sponsored attacks.
- Hold annual cyber defense exercises nationally.

## Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

| Done                     | Deliverable                             | Requirements                                                               |
|--------------------------|-----------------------------------------|----------------------------------------------------------------------------|
| <input type="checkbox"/> | <b>Project Report</b>                   | Minimum 25 pages, wellformatted, all 15 parts completed, diagrams included |
| <input type="checkbox"/> | <b>PowerPoint Presentation</b>          | 15 to 20 slides, covering all major sections of the law                    |
| <input type="checkbox"/> | <b>Flowchart: Cybercrime Reporting</b>  | Shows complete incident reporting process from detection to resolution     |
| <input type="checkbox"/> | <b>Flowchart: Investigation Process</b> | Shows NCSA investigation steps from complaint to judgment                  |

| Criteria                                                                                      | Max Marks  | Marks Obtained |
|-----------------------------------------------------------------------------------------------|------------|----------------|
| Research Quality — depth and accuracy of information, use of real-world examples              | 20         |                |
| Law Structure — completeness, logical organization, and legal clarity of the law              | 20         |                |
| Innovation — originality of solutions, creative approach to cybersecurity challenges          | 15         |                |
| Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures | 15         |                |
| Presentation — quality of the PowerPoint presentation and oral delivery                       | 10         |                |
| Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables         | 10         |                |
| Case Studies — realism, detail, and correct application of the law                            | 10         |                |
| <b>TOTAL</b>                                                                                  | <b>100</b> |                |

## Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

| Done | Deliverable                            | Requirements                                           |
|------|----------------------------------------|--------------------------------------------------------|
| [ ]  | <b>Flowchart: Data Breach Response</b> | Shows data breach response procedure for organizations |

|     |                                           |                                                                         |
|-----|-------------------------------------------|-------------------------------------------------------------------------|
| [ ] | <b>Infographic: Citizens' Rights</b>      | Visual summary of all citizens' digital rights                          |
| [ ] | <b>Infographic: Cybercrime Categories</b> | Visual taxonomy of all defined cybercrimes with penalties               |
| [ ] | <b>Infographic: Security Framework</b>    | Visual overview of the data protection and corporate security framework |

**Good Luck!**

This project challenges you to think like a lawmaker, a technologist, and a citizen. Research realworld cybersecurity laws (India's IT Act, GDPR, Budapest Convention) for inspiration. Be creative, be thorough, and make VIT's digital future secure!